$$\frac{\partial M}{\partial t} = D\nabla^2 M - \frac{M}{T_{2B}}, \qquad \mathbf{r} \in \Omega$$

$$D\,\mathbf{n}\cdot\nabla M+\rho_2 M=0,\qquad \mathbf{r}\in\partial\Omega$$

$$S(t) = \int_{\Omega} M(\mathbf{r}, t)\, \mathrm{d}\Omega, \qquad M(\mathbf{r}, 0) = M_0$$

$$M(\mathbf{r},t) = e^{-t/T_{2B}}\,\phi_n(\mathbf{r})\,e^{-t/T_{2S}}$$

$$\nabla^2\phi_n + \frac{\xi_n^2}{a^2}\phi_n = 0, \qquad D\,\mathbf{n}\cdot\nabla\phi_n + \rho_2\phi_n = 0$$

$$\kappa = \frac{\rho_2 \, a}{D}$$

$$\frac{1}{T_{2,n}} = \frac{1}{T_{2B}} + \frac{D\,\xi_n^2}{a^2}, \qquad \frac{S(t)}{S(0)} = \sum_n A_n\, e^{-t/T_{2,n}}$$

$$\phi_n(r) = J_0\left(\xi_n \frac{r}{R}\right)$$

$$\xi_n \frac{J_1(\xi_n)}{J_0(\xi_n)} = \kappa = \frac{\rho_2 R}{D}$$

$$A_n = \frac{4\kappa^2}{\xi_n^2\left(\xi_n^2 + \kappa^2\right)}$$

$$\phi_n(y) = \cos\left(\xi_n \frac{y}{h}\right)$$

$$\xi_n \tan \xi_n = \kappa = \frac{\rho_2 h}{D}$$

$$A_n = \frac{2\sin^2 \xi_n}{\xi_n \left(\xi_n + \sin \xi_n \cos \xi_n\right)}$$

$$\frac{J_1(x)}{J_0(x)} \approx \frac{x}{2} + \frac{x^3}{16} \implies \xi_0^2 \approx 2\kappa\left(1 - \frac{\kappa}{4}\right)$$

$$x \tan x \approx x^2 + \frac{x^4}{3} \implies \xi_0^2 \approx \kappa \left(1 - \frac{\kappa}{3}\right)$$

$$\boxed{\frac{1}{T_2} \approx \frac{1}{T_{2B}} + \rho_2 \frac{S}{V}}, \qquad \frac{S}{V} = \begin{cases} 2/R, & \text{二维圆盘} \\ 1/h, & \text{二维狭缝} \end{cases}$$

$$R = \frac{2\rho_2}{1/T_2 - 1/T_{2B}}, \qquad h = \frac{\rho_2}{1/T_2 - 1/T_{2B}}$$

$$\kappa = \frac{\rho_2 R}{D} = \frac{0.005 \times 20}{2.0} = 0.05 \ll 1$$

$$\xi_0 \frac{J_1(\xi_0)}{J_0(\xi_0)} = 0.05 \implies \xi_0 = 0.314262, \quad \xi_0^2 = 0.0987604$$

$$\frac{1}{T_{2S}} = \frac{D\xi_0^2}{R^2} = \frac{2.0 \times 0.0987604}{20^2} = 4.93802 \times 10^{-4}\ \text{ms}^{-1} \implies T_{2S} = 2025.10\ \text{ms}$$

$$\frac{1}{T_2} = 4.93802 \times 10^{-4} + 3.33333 \times 10^{-4} = 8.27135 \times 10^{-4}\ \text{ms}^{-1} \implies T_2 = 1208.99\ \text{ms}$$

$$A_0 = \frac{4\kappa^2}{\xi_0^2(\xi_0^2+\kappa^2)} = 0.99995, \qquad T_{2,1} = 13.47 \text{ ms}, \; A_1 = 4.6 \times 10^{-5}$$

$$\frac{1}{T_2} \approx \frac{2 \times 0.005}{20} + \frac{1}{3000} \implies T_2 \approx 1200.0 \text{ ms} \quad (-0.74\%)$$

$$\kappa = \frac{0.005 \times 5}{2.0} = 0.0125, \qquad \xi_0 = 0.157867$$

$$\frac{1}{T_{2S}} = \frac{2.0 \times 0.0249220}{5^2} = 1.99376 \times 10^{-3}\ \mathrm{ms}^{-1}$$

$$T_{2S} = 501.56 \text{ ms}, \qquad T_2 = \left(\frac{1}{501.56} + \frac{1}{3000} \right)^{-1} = 429.72 \text{ ms}$$

$$\frac{1}{T_2} \approx \frac{2 \times 0.005}{5} + \frac{1}{3000} \implies T_2 \approx 428.57 \text{ ms} \quad (-0.27\%)$$

$$\kappa = \frac{0.005 \times 0.05}{2.0} = 1.25 \times 10^{-4}, \qquad \xi_0 = 0.0111801$$

$$\frac{1}{T_{2S}} = \frac{2.0 \times 1.24995 \times 10^{-4}}{0.05^2} = 0.0999958 \text{ ms}^{-1}$$

$$T_{2S} = 10.0004 \text{ ms}, \qquad T_2 = 9.9672 \text{ ms}$$

$$\frac{1}{T_2} \approx \frac{0.005}{0.05} + \frac{1}{3000} \implies T_2 \approx 9.9668 \text{ ms} \quad (-0.004\%)$$

$$\frac{1}{T_2}=\frac{1}{T_{2B}}+\rho_2\frac{S}{V}\implies R=\frac{2\rho_2}{\dfrac{1}{T_2}-\dfrac{1}{T_{2B}}}\ (\text{圆盘}),\qquad h=\frac{\rho_2}{\dfrac{1}{T_2}-\dfrac{1}{T_{2B}}}\ (\text{狭缝})$$

$$R_B = \frac{2 \times 0.005}{1/1209.26 - 1/3000} = \frac{0.01}{4.93622 \times 10^{-4}} = 20.26\ \mu\text{m} \quad (+1.3\%)$$

$$R_S = \frac{2 \times 0.005}{1/429.83 - 1/3000} = \frac{0.01}{1.99315 \times 10^{-3}} = 5.017\ \mu\text{m} \quad (+0.3\%)$$

$$h = \frac{0.005}{1/10.03 - 1/3000} = \frac{0.005}{9.94047 \times 10^{-2}} = 0.0503\ \mu\text{m} \quad (+0.6\%)$$

$$R_B = \frac{2 \times 0.005}{1/1208.99 - 1/3000} = 20.25\ \mu\text{m}, \quad R_S = 5.016\ \mu\text{m}, \quad h = 0.0500\ \mu\text{m}$$